

# AI-Based Medical Image Diagnosis & Triage Support System

Chest X-ray (CheXNet-style) + Skin Lesion Analysis | Full Learning Roadmap & Resource Kit

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India faces a severe shortage of radiologists and dermatologists relative to its population, especially in rural and semi-urban areas, which delays diagnosis of conditions like pneumonia, TB, and skin cancer. This project builds an AI-assisted screening tool that flags high-risk cases from chest X-rays and dermatoscopic skin images, helping prioritize patients for specialist review. It is explicitly a **triage / decision-support tool, not a diagnostic replacement**.

## Why It's Credible

- **CheXNet (Stanford, 2017)** — a 121-layer DenseNet CNN that matched or exceeded radiologist performance on pneumonia detection from chest X-rays. [Paper \(arXiv:1711.05225\)](#)
- **Esteva et al., Nature (2017)** — a CNN trained on skin lesion images performed on par with board-certified dermatologists in classifying skin cancer. [Paper \(Nature\)](#)

Referencing these in your proposal shows the mentor this isn't a random idea — it is grounded in published, peer-reviewed research.

## Ethical Disclaimer (state this explicitly to your mentor)

This system is a **decision-support / triage tool**, not a diagnostic replacement for a licensed radiologist or dermatologist. It is trained exclusively on public, de-identified research datasets (NIH, ISIC, Stanford) — not real patient data — so there are no IRB or patient-consent issues at this stage.

## 1. Datasets

| Dataset                       | Size                              | Use                                   | Link                                    |
|-------------------------------|-----------------------------------|---------------------------------------|-----------------------------------------|
| NIH ChestX-ray14              | 112,120 images, 14 disease labels | Chest X-ray module (multi-label)      | <a href="#">NIH Box / Kaggle mirror</a> |
| ISIC Skin Cancer Archive      | 33,000+ labeled lesion images     | Skin lesion module                    | <a href="#">ISIC Archive</a>            |
| HAM10000 (ISIC 2018 subset)   | 10,015 images, 7 classes          | Easier starting point for skin module | <a href="#">Kaggle</a>                  |
| CheXpert (Stanford, optional) | 224,000+ images                   | Additional / validation set           | <a href="#">Stanford CheXpert</a>       |

*Tip: start with HAM10000 (smaller, cleaner) for your skin module MVP, then scale to full ISIC archive in Semester 2.*

## 2. Tech Stack

- **ML:** Python, PyTorch (recommended over TensorFlow for research-style repos — most CheXNet/Grad-CAM reference code is PyTorch), OpenCV, pytorch-grad-cam
- **Serving:** FastAPI (ML microservice)
- **Dashboard:** React + Express (MERN), MongoDB for metadata
- **Deployment:** Docker, optional logging/versioning layer

## 3. What Makes It Non-Trivial (say this explicitly to your mentor)

- Multi-label classification — a chest X-ray can show multiple conditions simultaneously, which is much harder than single-label classification.
- Class imbalance handling — rare diseases are underrepresented; needs weighted loss / oversampling.
- Explainability (Grad-CAM) is a hard, actively-researched CV problem, not a bolt-on feature.
- End-to-end deployable system with a real API and dashboard, not just a notebook.

## 4. Roadmap with Resources (8 Months, Split Across 2 Semesters)

### Months 1–2 — Literature Review, Dataset Exploration, Preprocessing Pipeline

**Goal:** Understand the CheXNet and Esteva papers, get comfortable with medical image data, build a clean preprocessing + augmentation pipeline and handle class imbalance.

- Paper: [CheXNet \(arXiv:1711.05225\)](#) — read this first, it's short and defines your whole Sem-1 target.
- Paper: [Esteva et al., Nature 2017](#) — skin lesion classification benchmark.
- YouTube playlist: [PyTorch Transfer Learning Tutorial](#) — ResNet/DenseNet fine-tuning basics.
- Official tutorial: [PyTorch Transfer Learning for Computer Vision](#)
- Read: [LearnOpenCV — Transfer Learning for Medical Images](#) (compares DenseNet/ResNet/EfficientNet on CheXpert — useful for your architecture-comparison section).
- Class imbalance in medical imaging: search "weighted BCE loss class imbalance pytorch" — use **weighted binary cross-entropy** for chest X-ray (multi-label) and **weighted CrossEntropy / oversampling** for skin lesions (HAM10000 is ~70% one class).

### Months 3–4 — Baseline Model Training + Evaluation (Semester 1 Submission)

**Goal:** Train two baseline models (chest X-ray multi-label, skin lesion 7-class), compare 2–3 backbone architectures, and evaluate properly with medically-relevant metrics.

- GitHub (chest X-ray, PyTorch CheXNet reimplementations — pick one as your starting scaffold):
  - – [arnoweng/CheXNet](#) — most-cited PyTorch reimplementation, reports AUROC ~0.847.
  - – [zoogzog/chexnet](#) — lets you swap DenseNet-121/169/201 easily, good for your architecture comparison.
  - – [gathierry/CheXNet-PyTorch](#) — clean train/val/test split matching the original paper.
- GitHub (skin lesion, HAM10000):
  - – [ishanrai05/skin-cancer-prediction](#) — PyTorch, ResNeXt101, ~88% val accuracy, good reference for pipeline structure.
  - – [zairza-cetb/HAM10000](#) — simpler CNN baseline, good for understanding the data format first.
  - – Browse more: [GitHub topic: ham10000](#) and [skin-lesion-classification](#)
- YouTube: [CampusX — FastAPI for Machine Learning \(playlist\)](#) (watch in parallel, prep for Sem 2 deployment).
- Metrics: implement AUC-ROC (per-class + macro), sensitivity/specificity, and confusion matrix. **Explicitly note to your mentor:** in medical AI, false negatives (missed disease) are far costlier than false positives, so optimize/report sensitivity at a chosen operating threshold, not just accuracy.

### Months 5–6 — Explainability (Grad-CAM) + Severity Scoring

**Goal:** Add Grad-CAM / Grad-CAM++ heatmaps so predictions are visually explainable, and move from flat labels to a risk/severity score.

- GitHub (use this directly — it's the standard library, don't reimplement from scratch): [jacobgil/pytorch-grad-cam](#) — supports GradCAM, GradCAM++, HiResCAM, works with CNNs and ViTs, includes ready notebooks.
- Tutorial: [Grad-CAM from Scratch with PyTorch Hooks \(Towards Data Science\)](#) — do this once manually so you understand the mechanism, even though you'll use the jacobgil library for the real pipeline.
- Reference: [Grad-CAM original paper \(arXiv:1610.02391\)](#)
- Paper repos already generate heatmaps — [berneylin/chexnet](#) has a ready `generate\_heatmap.py` you can study for chest X-ray CAM overlay logic.
- Severity scoring idea: instead of a flat label, combine the model's confidence score with the affected area from the Grad-CAM mask (e.g. % of lung field highlighted) to produce a Low/Medium/High risk tier — this is a

good "non-trivial" talking point for your mentor review.

## Months 7–8 — Dashboard + Deployment + Final Report/Demo (Semester 2 Submission)

**Goal:** Wrap the trained models behind a FastAPI ML microservice, build a doctor-facing MERN dashboard, and (optionally) containerize everything with Docker.

- YouTube: [CampusX — FastAPI for Machine Learning \(playlist\)](#) — Indian-audience, beginner-friendly, covers request/response schemas with Pydantic.
- YouTube: [Krish Naik — Deploy ML Models with FastAPI](#) (GitHub: [krishnaik06/FastAPI](#))
- YouTube: [Deploy ML models with FastAPI, Docker & Heroku \(AssemblyAI\)](#) — good for the Docker + deployment step.
- YouTube: [Deploy ML Models with FastAPI & Docker \(Full Tutorial\)](#)
- Architecture for you: React frontend → Express API (auth, patient/session metadata in MongoDB) → calls FastAPI ML microservice (image in, JSON with prediction + confidence + base64 heatmap out) → render heatmap overlay on the uploaded image in React.
- Optional MLOps layer: Dockerize the FastAPI service, add basic request logging and a model-version tag in responses — this alone is worth a lot of "production-readiness" credit with a mentor.

## 5. Curated Resource Quick-List

### Papers

- [CheXNet paper](#)
- [Esteva et al., Nature 2017 \(skin cancer\)](#)
- [Grad-CAM original paper](#)

### GitHub Repos

- [arnoweng/CheXNet](#) — chest X-ray, PyTorch, well-documented
- [zoogzog/chexnet](#) — multiple DenseNet variants + heatmap generator
- [gathierry/CheXNet-PyTorch](#) — clean data pipeline
- [ishanrai05/skin-cancer-prediction](#) — HAM10000, PyTorch
- [zairza-cetb/HAM10000](#) — beginner-friendly HAM10000 baseline
- [jacobgil/pytorch-grad-cam](#) — explainability, use directly in your pipeline
- [krishnaik06/FastAPI](#) — FastAPI deployment reference

### YouTube

- [PyTorch Transfer Learning Tutorial](#)
- [CampusX — FastAPI for Machine Learning \(playlist\)](#)
- [Krish Naik — Deploy ML Models Using FastAPI](#)
- [Deploy ML models with FastAPI, Docker, Heroku \(AssemblyAI\)](#)
- [Deploy ML Models with FastAPI & Docker \(Full Tutorial\)](#)

### Datasets

- [NIH ChestX-ray14 \(Kaggle mirror\)](#)
- [ISIC Archive](#)
- [HAM10000 \(Kaggle\)](#)
- [CheXpert \(Stanford\)](#)

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*Bhai, suggestion: start Month 1 by cloning zairza-cetb/HAM10000 and arnoweng/CheXNet locally, run their pretrained/demo scripts on a small subset first — getting a working end-to-end pipeline (even a bad one) in week 1 beats reading papers for a month before writing any code.*